

SANIDHYA VARDHAN SINGH

Software Engineer — Backend Systems, Data Engineering & Generative AI

#57 Kucha Kanshi Ram, Shah Jain Bazaar, Jammu, Jammu & Kashmir, India

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PROFESSIONAL SUMMARY

Software Engineer with hands-on experience in backend development, distributed systems, and large-scale data processing pipelines using Python, SQL, Java, and Go. Proven ability building production-oriented Generative AI and multi-agent systems on Microsoft Azure AI Foundry, validated by 19 completed Microsoft Learn credentials spanning AI fundamentals, natural language processing, computer vision, and agentic AI orchestration. Skilled in designing REST APIs, optimizing SQL databases, and deploying scalable pipelines with Apache Kafka and Apache Spark across AWS, Azure, and GCP. Additional practical exposure to cybersecurity network defense, threat mitigation, and forensic data analytics.

CORE TECHNICAL SKILLS

- Programming Languages: Python, SQL, Java, Go, C, C++, Rust
- Artificial Intelligence & Machine Learning: Generative AI, Multi-Agent Systems, Agentic AI Orchestration, Natural Language Processing (NLP), Computer Vision, TensorFlow, PyTorch, scikit-learn
- Microsoft Azure AI Stack: Microsoft Azure AI Foundry, Microsoft Agent Framework, Foundry IQ, Model Context Protocol (MCP), Azure AI Agents (A2A)
- Backend & Data Engineering: REST API Design, Apache Kafka, Apache Spark, PostgreSQL, MongoDB, Redis
- Cloud & DevOps: AWS, Microsoft Azure, Google Cloud Platform (GCP), Docker, Kubernetes, CI/CD, Git
- Security & Analytics: Network Access Control, Threat Mitigation, Forensic Data Analytics, System Log Analysis

WORK EXPERIENCE

Backend Web Developer — G Caffe

Noida, India | May 2025 – Aug 2025

- Engineered and deployed scalable backend services for enterprise web applications using Python, Java, and optimized SQL schemas.
- Implemented automated data processing routines and unit/integration testing frameworks, ensuring high platform availability and zero-loss data ingestion.
- Partnered with senior engineers in cross-functional architecture reviews to align backend capabilities with analytics requirements.
- Used Git for version control and integrated CI/CD pipelines to streamline automated testing and containerized deployments on AWS.

Website Developer — One Yes Info Tech Solutions

Chennai, India | Dec 2023

- Built metric-focused application features emphasizing processing integrity and minimalist, scalable software architecture.
- Integrated REST APIs and JSON data formats to enable high-throughput communication between frontend and backend systems.
- Tuned database queries and reduced application latency using in-memory caching with Redis.
- Maintained strict code documentation standards and high architectural compliance.

Corporate Job Simulation Participant — Deloitte (via Forage)

Remote | Jun 2026

- Cybersecurity Simulation: Performed network access control defense, system log analysis, and corporate threat mitigation tasks; presented findings to stakeholders.
- Data Analytics Simulation: Conducted forensic technology investigations, identified dataset anomalies, and delivered corporate metadata insights.
- Technology Simulation: Completed production-grade coding exercises, optimizing backend workflows and software engineering deliverables.

PROJECTS

AI-Driven Wildlife Deterrence System | Dec 2024 – Feb 2025

- Architected an end-to-end distributed system in Python and TensorFlow using computer vision to process optical feeds and classify wildlife patterns.
- Built streaming data-ingestion mechanisms handling concurrent sensor payloads with sub-second processing latency.
- Containerized microservices with Docker and deployed to Google Cloud Platform (GCP) for scalable operation.

Distributed Systems & Open-Source Contributions | Present

- Developed high-performance distributed systems experiments exploring memory safety and transactional execution speed using Go and Java.
- Contributed to open-source frameworks, mapping functional programming patterns and low-level systems architecture.

E-Vehicle Thermal Management Solution

- Designed a high-efficiency closed-loop liquid cooling solution for electric vehicle battery, motor, and inverter systems.
- Modeled system thermodynamics and mapped multi-point dependencies across the powertrain.

MICROSOFT AZURE AI CERTIFICATIONS (19 CREDENTIALS — MICROSOFT LEARN, JUNE 2026)

AI Fundamentals

- Introduction to AI Concepts
- Introduction to Computer Vision Concepts
- Introduction to AI Speech Concepts
- Introduction to Natural Language Processing Concepts
- Introduction to AI-Powered Information Extraction Concepts
- Introduction to Generative AI and Agents

Generative AI Development on Microsoft Foundry

- Plan and Prepare to Develop AI Solutions on Azure
- Select, Deploy, and Evaluate Microsoft Foundry Models
- Develop a Generative AI Chat App with Microsoft Foundry
- Develop Generative AI Apps That Use Tools
- Optimize Generative AI Model Performance with Microsoft Foundry
- Implement a Responsible Generative AI Solution in Microsoft Foundry

Agentic AI & Multi-Agent Systems

- Discover Azure AI Agents with A2A
- Orchestrate a Multi-Agent Solution Using the Microsoft Agent Framework
- Develop an AI Agent with Microsoft Agent Framework
- Build Agent-Driven Workflows Using Microsoft Foundry
- Integrate Your Agent with Microsoft 365
- Build Knowledge-Enhanced AI Agents with Foundry IQ
- Integrate MCP Tools with Azure AI Agents

ADDITIONAL CERTIFICATIONS

- Advanced AI & Agentic Frameworks
- Professional Data Science & Analytics Certification
- MERN Stack Development
- Full Stack Programming
- Scrum Master

EDUCATION

Bachelor of Engineering, Computer Science Engineering — Velammal Engineering CollegeIndia | Nov 2021 – Apr 2025

Higher Secondary (12th Grade) — K.C. International School

India | May 2020 – May 2021

ACHIEVEMENTS

- PMSSS Scholarship — All India Council for Technical Education (AICTE)
- Governor's Award
- NCAIF'24 — ACM 9th National Conference on Artificial Intelligence (Participant/Attendee), 2024